

Programming in C

Lab 1: IDE for C coding

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Spring Semester 2022

Outline

- 1 Codeblocks IDE
- 2 Visual Studio Code
- 3 Variables

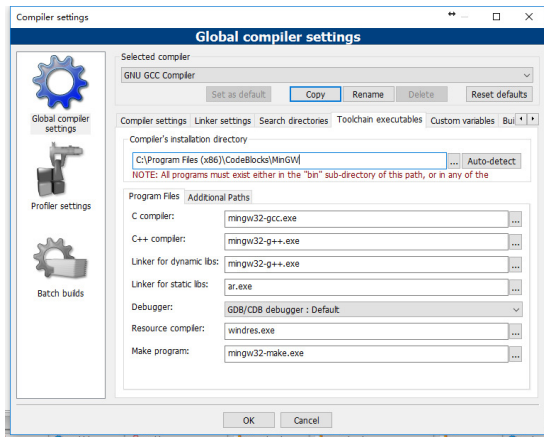
Codeblocks

- It is free software, available at following link

[CodeBlocks 16.01 Windows binary](#)

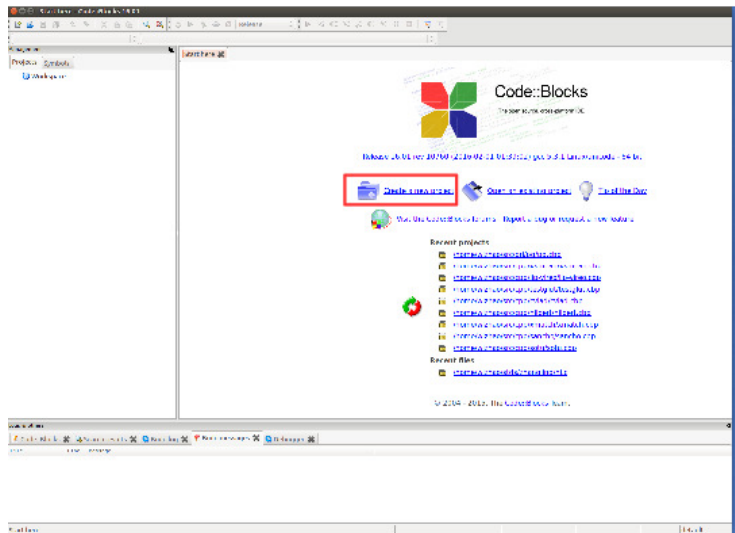
- Lightweight and stable
- Cross platform: Linux, Windows and MacOS

Compiler Setup



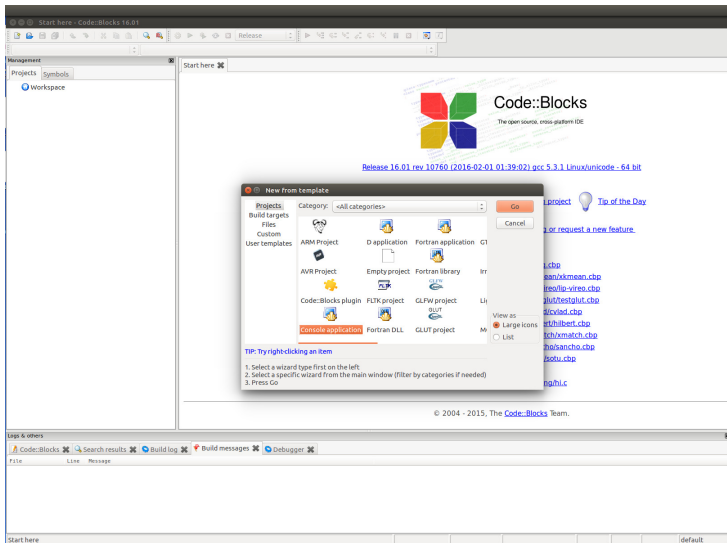
- 1 Go to menu: Settings→Compiler Settings→**Toolchain Executables**
- 2 Type in the path for mingw compiler, e.g., C:/Program Files (x86)/CodeBlocks/MingW/

Create a project: step 1



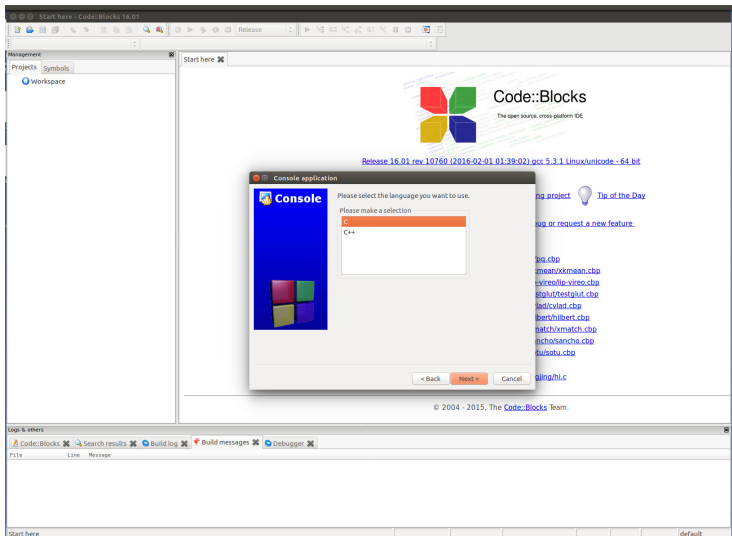
- Create/New a C project

Create a project: step 2



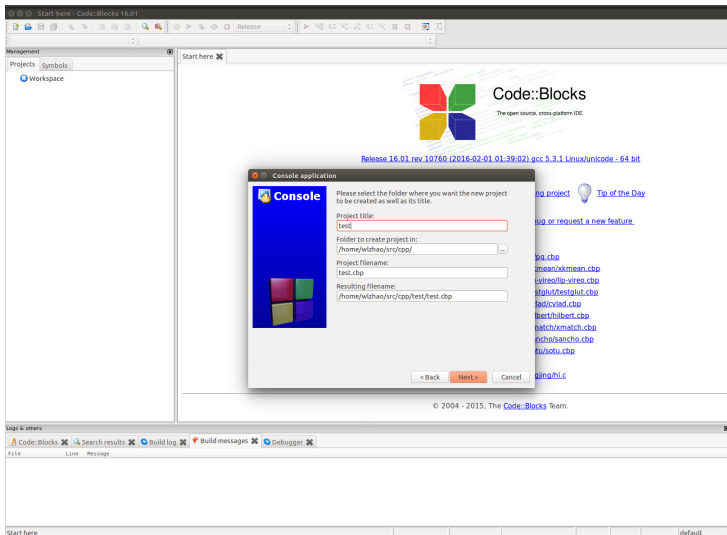
- Choose project type “Console application”

Create a project: step 3



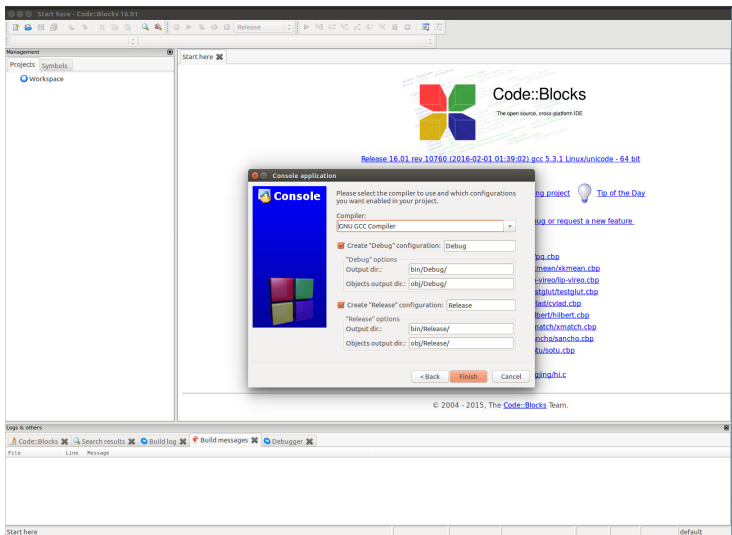
- Set it as a “C” project

Create a project: step 4



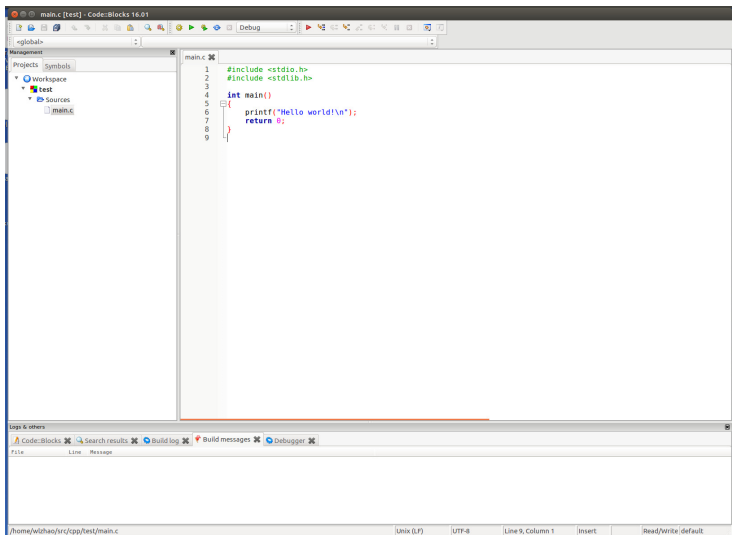
- Give a name for your project

Create a project: step 5



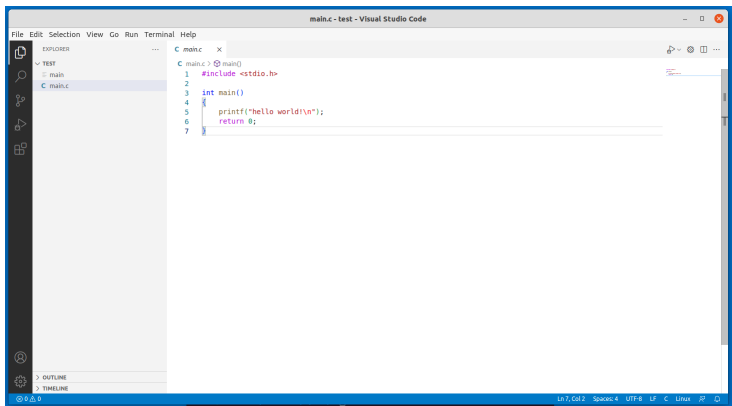
- Choose “C” compiler for your project

Create a project: step 6



- Start to work with your project

Main interface of VS code



- VS code is the most powerful and convenient Editor¹
- One editor for designed for various programming languages, C, C++, Python and Java, etc

¹<https://code.visualstudio.com/download>

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Variable Types and Operations

- Try following code:

```
1 #include <stdio.h>
2 int main( )
3 {
4     int a = 5, b = 3;
5     float val1 = a/b;
6     float val2 = (float)(a/b);
7     float val3 = (float)(a+0.0)/b;
8     printf("a/b is: -val1 ==-%f\n", val1);
9     printf("a/b is: -val2 ==-%f\n", val2);
10    printf("a/b is: -val3 ==-%f\n", val3);
11 }
```

Character and String

- Try following code:

```
1 #include <stdio.h>
2 int main()
3 {
4     char *str1 = "abc\0de";
5     char *str2 = "abc\rde";
6     char *str3 = "abc\tde";
7     char *str4 = "abc\t\b\b\b\b\bde";
8     char ch1 = 'A'+6;
9     char ch2 = 'A' + 'B';
10    printf("%s\n", str1);
11    printf("%s\n", str2);
12    printf("%s\n", str3);
13    printf("%s\n", str4);
14    printf("%c, - value=%d\n", ch1, ch1);
15    printf("%c, - value=%d\n", ch2, ch2);
16 }
```

How Variable Behaves

- Guess the values of **a**, **i** and **j**
- Try following code to verify your answer:

```
1 #include <stdio.h>
2 int main()
3 {
4     int i = 4, j = 6;
5     int a = i + j;
6     printf("a == %d\n", a);
7     i = j - i;
8     j = i + j;
9     printf("i == %d, j == %d\n", i, j);
10 }
```

printf() and Precision Control

- Given float number $a = 231.36952$, integer number $b = 39$ and integer number $c=0xEE$
 - Print out a with 2 digits precision and 3 digits precision respectively
 - Print out octal and hexadecimal values of b
 - Print out decimal and octal number of c

```
1 #include <stdio.h>
2 int main( )
3 {
4     float a = 231.36952;
5     int b = 39;
6     int c = 0xEE;
7 }
```

Print out the values of different variables

- Given following variables have been defined
- Please show the number of bytes they occupy in the memory

```
1 #include <stdio.h>
2 int main()
3 {
4     char ch = 'B';
5     int a = 0;
6     short b = 1024;
7     double c = 0.1;
8     float d = 22;
9     double e = 3.1415926;
10 }
```

- Please print the values of different variables on the screen
- For example,

```
1 printf("%f\n", d);
```

Precision of `float` and `double`

```
1 #include <stdio.h>
2 int main()
3 {
4     double c = 0.1;
5     float d = 22;
6     double e = 3.1415926;
7     printf("e==%1.4lf\n", e);
8     printf("e==%1.3lf\n", e);
9     printf("e==%1.2lf\n", e);
10 }
```

Print integer numbers with formatting

- Print out a=234, b=5, c=123, d=55, two numbers in each line
- Numbers are separated by ','
- Each number occupy 6 digital position, right-aligned

```
__ __ __ 234, __ __ __ __ 5
__ __ __ 123, __ __ __ __ 55
```

Print integer numbers with right-aligned

- Try following code:

```
1 #include <stdio.h>
2 int main()
3 {
4     int a = 234;
5     int b = 5;
6     int c = 1231;
7     int d = 55;
8     printf("%6d, -%6d\n", a, b);
9     printf("%6d, -%6d\n", c, d);
10 }
```

Print integer numbers with left-aligned

- Try following code:

```
1 #include <stdio.h>
2 int main()
3 {
4     int a = 234;
5     int b = 5;
6     int c = 1231;
7     int d = 55;
8     printf("%-6d, %-6d\n", a, b);
9     printf("%-6d, %-6d\n", c, d);
10 }
```